Forecasting Recall Feasibility Before Calibration: A Held-Out Diagnostic Study of Capped IVF Retrieval

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Abstract. Hard retrieval budgets and limited calibration evidence create different obstacles to recall certification. We test whether centroid geometry helps distinguish these obstacles before collecting calibration labels. Matched tree models predict maximal-prefix recall losses using design labels and either occupancy features or additional geometry. Frozen five-seed pilots trigger confirmation on separate queries from GloVe and COCO CLIP retrieval. Geometry reduces severity prediction error, but does not confer a distribution-free safety guarantee. Operational gains vary across workloads; source-level lookahead costs also vary across seeds. We release protocols, disjoint query roles, negative findings, and exact traces while retaining limits on novelty and generalization.

1 A diagnostic question, not a new calibration theorem

Recall calibration is useful only when a search policy can satisfy the requested risk at the permitted work budget. ConANN already provides adaptive IVF recall calibration [1], and LTT/Pareto Testing already address risk-constrained configuration selection [2, 3]. Feasibility diagnostics also have prior work outside retrieval [4]. Merely applying these tools to another loss is not our claimed contribution.

We investigate a narrower empirical question: *does design-trained query geometry improve predictions of the best recall attainable by a fixed capped prefix class, beyond cap and occupancy information alone?* Such predictions could support triage before expensive additional labeling, but remain fallible forecasts. We separately examine actual risk certification and executed-work accounting; prediction accuracy cannot substitute for either.

The study extends earlier unsuccessful adaptive-calibration experiments. We freeze the protocol, add cross-modal retrieval and evaluate fresh queries. It confirms one prespecified diagnostic criterion, retains a failed accounting gate, and makes no claim of a new statistical algorithm or optimal search policy.

2 Retrieval floors, evidence, and forecast targets

Fix $N = 10{,}000$ corpus vectors, an IVF index with $M = 64$ lists ordered by centroid distance, and $k = 10$. Lists are scanned completely. For exact neighbors $T_k(q)$ and candidate union $A_p(q)$, define

$$L(q,p) = 1 - \frac{|T_k(q) \cap A_p(q)|}{k}, \qquad C(q,p) = 64 + \sum_{i=1}^{p} |I_i(q)|. \tag{1}$$

The cap is $B = 64 + \lfloor bN \rfloor$, with $b \in \{.05, .10, .25, .50, 1\}$. Empty prefixes have loss 1. Every admissible whole-list prefix policy obeys $\pi(q) \le p_B(q)$, where

$$p_B(q) = \max\{p : C(q,p) \le B\}, \qquad L_B(q) = L(q, p_B(q)). \tag{2}$$

Nested candidates imply $L(q,\pi(q)) \ge L_B(q)$ pointwise, including for nondecreasing severity transforms and conditional averages. This elementary floor is class-specific; it is not a knapsack hardness theorem or an impossibility result for graphs, partial lists or alternative probe orders.

Each diagnostic predicts the three bounded targets

$$Z_B(q) = \big(L_B(q),\ \mathbf{1}\{L_B(q) > .20\},\ \mathbf{1}\{L_B(q) > .50\}\big). \tag{3}$$

Four fixed overlapping groups comprise all queries and the upper 20%, 10% and 5% of design-defined centroid ambiguity. A workload/group regime is empirically feasible if its three average targets are at most $(.10, .10, .05)$. We call a diagnostic prediction an *admission* only when all three predicted averages meet these thresholds. It is not a certificate.

Certification uses standard fixed-sequence LTT: the maximum of 12 one-sided Chernoff p-values at .025. For constraint j with target a_j, group count n_j and bounded empirical risk $\widehat{R}_j$, the test is

$$p_j = \begin{cases} \exp\{-n_j \,\mathrm{kl}(\widehat{R}_j \| a_j)\}, & \widehat{R}_j < a_j,\ n_j > 0, \\ 1, & \text{otherwise.} \end{cases} \tag{4}$$

Here kl is binary relative entropy. Confidence is per fixed method/cap, not simultaneous over the displayed grid. Prospective safety requires a design-fixed path and a matching within-group query law; these source-law assumptions are not replaced by a predictor's accuracy. A reverse floor test at .025/12 can supply an infeasibility witness; otherwise inability to certify is *unknown*. Zero events for a .05 severity target require at least $\lceil \log(.025)/\log(.95) \rceil = 72$ group examples under this test. This is a count requirement of the implemented procedure, not a universal lower bound or an explanation for every unknown. More evidence can reveal infeasibility instead of producing a certificate.

3 Frozen pilots and confirmation protocol

We use normalized GloVe-25 embeddings [5] and author-released COCO text-to-image CLIP ViT-B/16 vectors [6, 7]. The latter has 512 dimensions, image corpus vectors and caption queries. Each workload uses five corpus/index seeds, 0–4. All runs share fixed query-role assignments: 500 design, 2,000 calibration, 500 pilot and 2,000 confirmation queries. The 24,771 GloVe training queries inspected in previous calibration studies were excluded before assigning new roles; previous official final TEST queries were never loaded. Recall concerns exact embedding neighbors, not semantic relevance judgments or downstream answer quality.

Both diagnostics are standard ExtraTrees regressors [8] with 64 trees. The occupancy control uses cap fraction, feasible prefix fraction, used corpus fraction and next-list fraction. Geometry adds centroid ambiguity, scaled nearest centroid distance, distance spread and two list-radius ratios. These features use no query's exact neighbors at prediction time; training still uses design labels. Both models receive the same four depth/leaf settings, selected on 300/200 design queries before refitting on all 500. Query identities are separated before expansion across caps.

H1 requires at least 10% severity Brier-error reduction over occupancy in at least 4/5 seeds on both datasets, nonzero admissions and at most 10% empirical false admissions among admitted regimes in each passing seed. The Brier score averages squared errors for the two severity indicators over queries and caps. Pilots passed in 5/5 GloVe and 4/5 COCO seeds. Source, protocol, pilot outputs and selected settings were then hashed before confirmation outcomes.

An input-only audit found repeated COCO vectors under different row IDs. Before confirmation labeling, six repeated vectors were replaced using a fixed unused-row order. Confirmation vectors are unique and disjoint in value from design, calibration and pilot roles. The small pilot overlap is disclosed, with a duplicate-excluded score sensitivity; no model was retuned.

Data	Occupancy	Geometry	Gain (%)	Gate
GloVe	0.0578 ± 0.0024	0.0469 ± 0.0014	18.8 ± 1.9	5/5
COCO CLIP	0.0949 ± 0.0034	0.0804 ± 0.0029	15.2 ± 2.6	4/5

Table 1: Fresh confirmation: Brier error and relative gain, mean ± seed SD. Gate counts repeat the unchanged full H1 criterion. Five seeds share query pools; they are not independent external datasets.

Separately, post-confirmation controls used only the old pilot, without retuning. Shuffling geometry raised mean Brier error from .0504 to .0618 on GloVe and from .0823 to .0964 on COCO. This supports predictive information beyond merely adding feature dimensions, but mechanistic attribution remains exploratory.

4 Decision consequences and certification boundaries

Geometry improves mean prediction error by 18.8% on GloVe and 15.2% on COCO. The confirmation criterion passes in 5/5 and 4/5 seeds. COCO seed 0 improves Brier error by 12.2% yet fails because one of nine admissions is false (11.1%). We retain this failure rather than relax the threshold.

Data	Diagnostic	Admit	False admit	False refuse	Correct
GloVe	Occupancy	60	0	0	100
GloVe	Geometry	60	0	0	100
COCO CLIP	Occupancy	52	10	0	90
COCO CLIP	Geometry	43	1	0	99

Table 2: Confirmation decisions over five caps, four groups and five index seeds: 100 *dependent* regimes per dataset. False admission means predicted feasible but empirically infeasible. False refusal is the converse. No binomial-trial or population error-bound interpretation is justified by these totals.

On GloVe, lower prediction error does not change decisions: the simpler control already classifies all evaluated regimes correctly. On COCO, geometry avoids nine false admissions without increasing false refusals in this workload. This is an observed engineering distinction, not a guaranteed reduction in deployment failures or labeling cost.

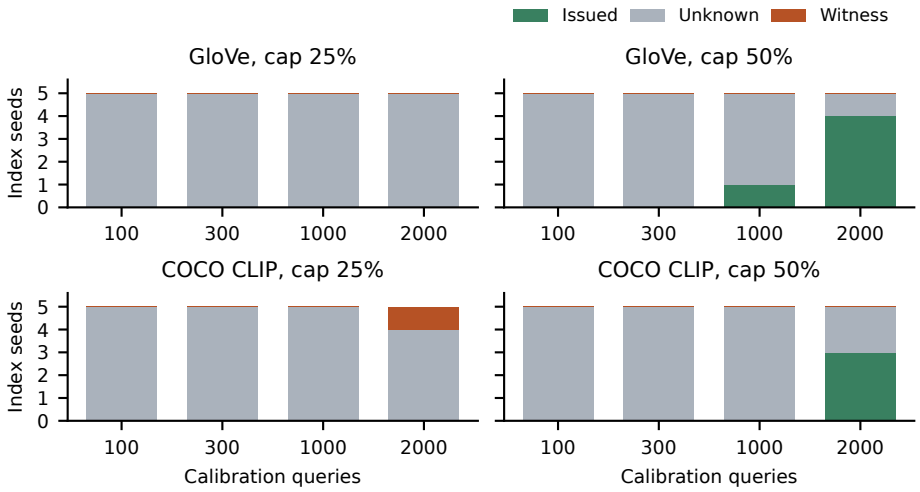


Fig. 1: Uniform-LTT status by calibration count and cap, all five index seeds. Unknown is distinct from a calibrated infeasibility witness. More labels can support a witness, not just issuance. The same labels are shared across methods and caps; labeling costs cannot be multiplied by the number of displayed cases.

5 Robustness, accounting audit and scope of evidence

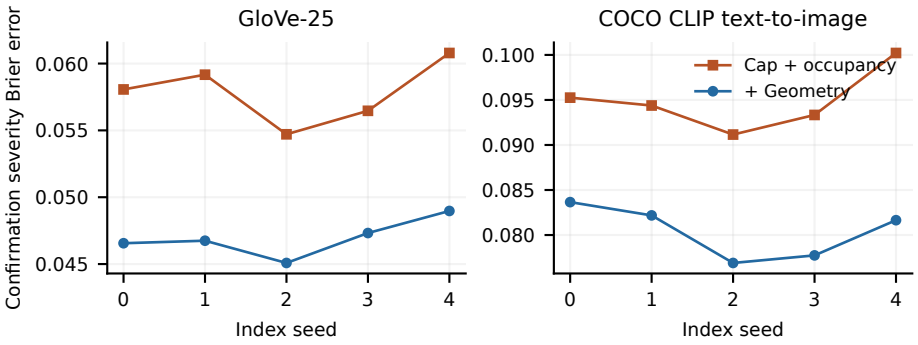


Fig. 2: Fresh-query Brier error for each index realization. Both diagnostics use identical design labels and tuning budgets. A better forecast is not necessarily a better admission decision, as GloVe demonstrates.

All ten index runs were retained in both phases. Each phase has 600 cases spanning caps, calibration counts and methods, plus 2,400 group rows. Checks rebuilt group assignments and verified raw/summary agreement, hard caps, score averages and returned/executed work. Paired query bootstraps use 2,000 resamples, keeping each query's caps together. They quantify conditional variation on these shared datasets, not independent-study uncertainty or universal safety.

Uniform, design-cell-shift and author-score paths are additionally calibrated under identical LTT group/tail constraints, with calibration counts 100, 300, 1000 and 2000. At the primary 25% cap no such path issued; an unissued diagnostic output is never reported as a certified deployed policy. The forecast comparison therefore does not claim to improve certified retrieval latency.

H2 separately tested executed versus returned-prefix work. In the pinned ConANN source, an extra list may be scanned before the previous result is restored. We compiled the exact scalar stop expressions and passed 10,000 differential checks plus live synthetic retrieval controls. The prespecified material-overhead pilot gate passed only 3/5 GloVe seeds versus 5/5 COCO seeds, hence failed across workloads. We do not rescue that hypothesis using confirmation statistics.

Crucially, the author's native distance counter already counts executed scans; this is *not* evidence that published ConANN latency is wrong. Our portable score adapter uses discrete CRC selection rather than GSL root solving, a fixed normalized-distance scale and locally implemented IVF. Isolated numerical dependencies resolved configuration, but the unmodified full engine fails on this Windows/MinGW toolchain at a POSIX allocation call. No full author-code performance reproduction is claimed.

6 Conclusions and remaining publication limits

The confirmed result is scoped: adding geometry to a matched design-trained occupancy predictor improves severity-risk forecasts on two retrieval workloads. It changes admission decisions on COCO, while GloVe shows that score improvements alone need not change an operational choice. Retrieval feasibility, statistical evidence and executed compute must remain separate quantities.

This is an empirical diagnostic contribution candidate, not a new conformal algorithm, formal admission guarantee or original impossibility theorem. The same two public query sources are reused across index seeds. Corpus size is 10,000, models are fixed, and semantic/OOD shift is untested. The rarest groups still have limited observations. Larger corpora, additional encoders, full ConANN/Faiss validation and a measured end-to-end diagnostic deployment remain necessary for broader claims. A large recorded wall-clock outlier is retained; timings on the loaded host do not establish hardware deadlines. No acceptance or exhaustive novelty claim is made.

Frozen code, role IDs, hashes, all outcomes, bootstrap intervals and attempt ledgers accompany this draft. Earlier failures are retained separately. Fresh confirmation was opened only after the pilot gate; prior official reserved TEST queries and thousands of unused COCO queries remain unexamined.

References

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